Decision Tree Regressor

Jab output ek number ho (jaise salary), tab Decision Tree kaise split karta hai — Variance Reduction ke saath.

1. Introduction — Regression Me Decision Tree

Pichhle notes me humne **Decision Tree Classifier** padha, jahan output ek **category** (Yes/No) hota tha. Ab dekhte hain jab output ek **continuous number** ho — jaise **Salary**, House Price, Temperature. Tab hum **Decision Tree Regressor** use karte hain. Tree ka structure wahi rehta hai (root → decision nodes → leaf nodes) aur algorithm bhi wahi **CART** hai (teacher ne tree ke saath CART likha hai).

Lekin ek badi dikkat aati hai. Classification me hum split ki purity **Entropy, Gini** aur feature select karne ke liye **Information Gain** se naapte the (notes ke top-right me ye teeno likhe hain). Ye teeno **categories** ki ginti par chalte hain, jabki yahan output numbers hain, jaise 40K, 42K, 52K... Isliye humein ek aisa measure chahiye jo **numbers ke bikhraav (spread)** ko naape. Wo hai **Variance**.

Point	Classification Tree	Regression Tree
Output	Category (Yes / No)	Number (jaise Salary)
Node ki purity	Entropy / Gini Impurity	**Variance** (Mean Squared Error)
Best split chunna	Information Gain	**Variance Reduction**
Leaf ka answer	Majority class	**Leaf ke values ka average (mean)**
sklearn class	DecisionTreeClassifier	DecisionTreeRegressor

2. Dataset

Teacher ka dataset: do input features aur ek output. **Experience (Exp)** numerical feature hai (saalon me), **Career Gap** categorical hai (Yes/No), aur output **Salary** hai (K = hazaar).

Exp (saal)	Career Gap	Salary (O/P)
2	Yes	40K
2.5	Yes	42K
3	No	52K
4	No	60K
4.5	Yes	56K

Pure data ka average salary: (40 + 42 + 52 + 60 + 56) / 5 = 250 / 5 = **50K**. Teacher ke example me splitting sirf **Exp** feature par ki gayi hai. Exp numerical hai, isliye pichhle notes wala **threshold** idea lagega: "Exp ≤ kuch value?" ka Yes/No sawal.

3. Variance — Regression Me Purity Naapne Ka Tareeka

Classification me pure node ka matlab tha "sirf ek class". Regression me pure node ka matlab hai **saare values ek-dusre ke bahut paas** (kam bikhraav). Bikhraav naapne ke liye **Variance** use hota hai:

$$\text{Variance} = (1/n) \times \Sigma\, (y - \bar{y})^2$$

Yahan **y** har sample ki salary hai aur **ȳ** unka **average** hai. Teacher ne ise **Mean Squared Error (MSE)** bhi likha hai, kyunki ye bhi "average se doori ka square, uska average" hi hai. Variance **kam** = values paas-paas (acha, pure); Variance **zyada** = values bikhre hue.

Root node ka variance (poore 5 salaries, mean = 50):

y (Salary)	$y - \bar{y}$ ($\bar{y} = 50$)	$(y - \bar{y})^2$
40	−10	100
42	−8	64
52	2	4
60	10	100
56	6	36
Total		**304**

$$\text{Variance(Root)} = 304 / 5 = 60.8$$

> **Yaad rakhein:** Regression me node ka "gandagi score" = uska **variance**. Classification me entropy/gini jaisa kaam yahi karta hai.

4. Variance Reduction — Best Split Kaise Chunein?

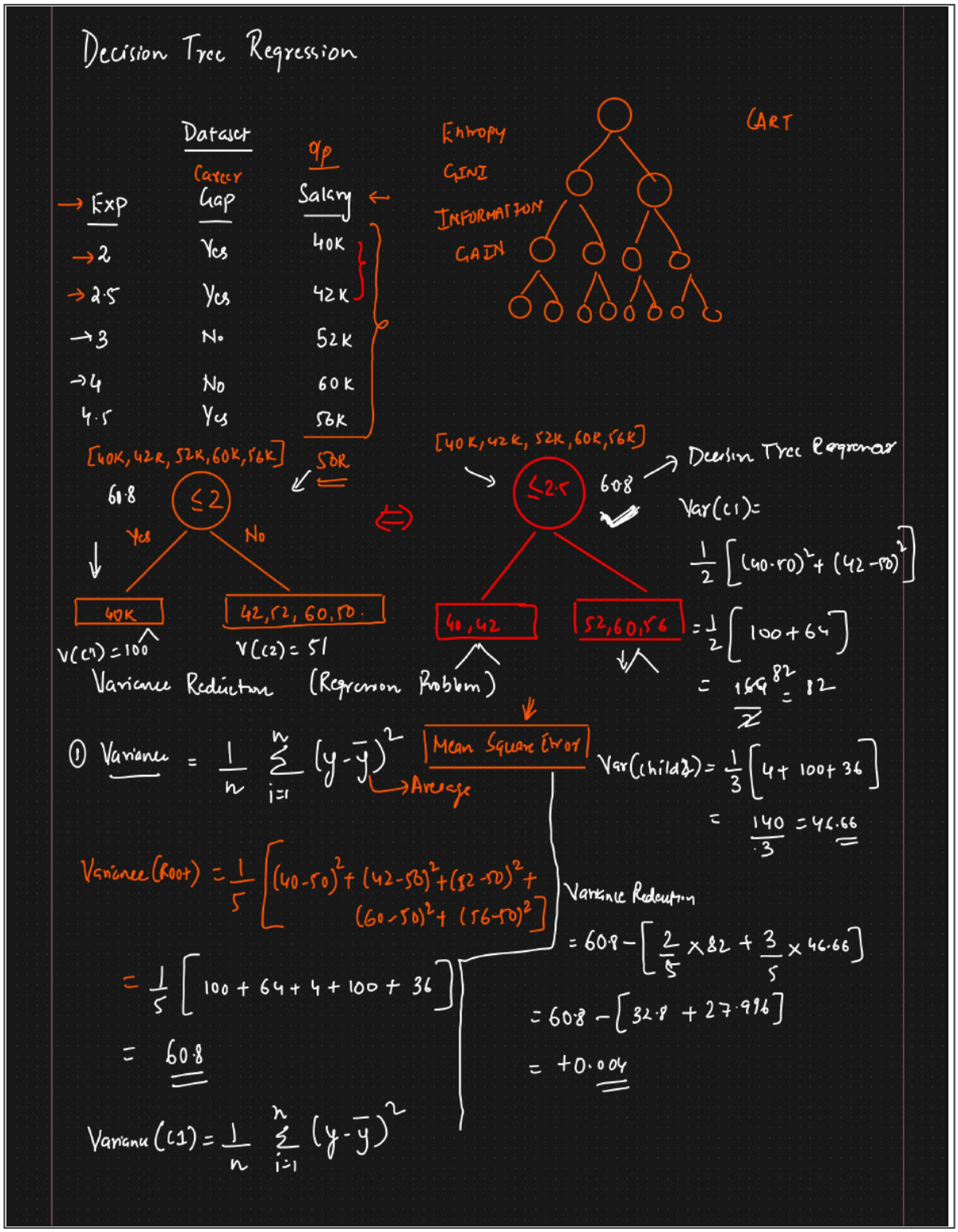
Ye bilkul Information Gain jaisa idea hai. Kisi split ke baad **parent ka variance** aur **children ke variance ka weighted average** compare karte hain. Jitna zyada variance ghate, utna achha split:

$$\text{Variance Reduction} = \text{Var(Root)} - \Sigma\, w_i \times \text{Var}(\text{Child}_i)$$

Symbol	Matlab
Var(Root)	Parent node ka variance (split se pehle)
$\text{Var}(\text{Child}_i)$	i-th child node ka variance
w_i	Weight = child me samples / parent me samples (jaise 2/5, 3/5)

> **Dhyaan dein:** Jis split ka **Variance Reduction sabse zyada** ho, wahi select hota hai. Ye regression ka "Information Gain" hai (classification me entropy ghatti thi, yahan variance ghatta hai).

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Isme dikhaya gaya hai: Salary dataset (Exp, Career Gap), CART tree ke saath Entropy/Gini/Information Gain, do candidate splits (Exp ≤ 2 aur Exp ≤ 2.5), Variance ka formula (MSE), Var(Root) = 60.8, aur dono splits ke Variance Reduction ki calculation.

5. Worked Example — Do Splits Ka Comparison

Teacher ne do candidate splits ke Variance Reduction compare kiye hain. Pehle **notes ke mutabik** calculation dekhte hain, phir ek zaroori correction.

Split A — Exp ≤ 2

Yes branch: **[40K]**. No branch: **[42, 52, 60, 56]**. Teacher ne dono ka variance 50 (root ka mean) se nikala hai:

$$\text{Var}(C_1) = (1/1) \times (40 - 50)^2 = 100$$
$$\text{Var}(C_2) = (1/4) \times [\ 64 + 4 + 100 + 36\ ] = 204 / 4 = 51$$

$$\text{VR} = 60.8 - [\ (1/5) \times 100 + (4/5) \times 51\ ] = 60.8 - (20 + 40.8) = 0$$

Split B — Exp ≤ 2.5

Yes branch: **[40K, 42K]**. No branch: **[52K, 60K, 56K]**.

$$\text{Var}(C_1) = (1/2) \times [\ (40 - 50)^2 + (42 - 50)^2\ ] = (100 + 64) / 2 = 82$$
$$\text{Var}(C_2) = (1/3) \times [\ 4 + 100 + 36\ ] = 140 / 3 = 46.66$$

$$\text{VR} = 60.8 - [\ (2/5) \times 82 + (3/5) \times 46.66\ ] = 60.8 - (32.8 + 27.996) = +0.004$$

Split	Yes branch	No branch	VR (notes me)
A: Exp ≤ 2	[40]	[42, 52, 60, 56]	0
B: Exp ≤ 2.5	[40, 42]	[52, 60, 56]	0.004

Teacher ka conclusion: **VR(Left split, ≤ 2) < VR(Right split, ≤ 2.5)**, isliye **Exp ≤ 2.5** wala tree chuna gaya (notes me tick lagaya hai).

5.1 Zaroori Correction — Child Ka Apna Mean Use Karo

Dhyaan dein — notes me ek concept ki galti: Teacher ne children ka variance bhi **root ke mean (50)** se nikala hai. Lekin kisi node ka variance hamesha **us node ke apne average** se nikalna chahiye. Root ke mean se nikalne par Variance Reduction hamesha **0** aata hai (kyunki weighted average wapas root ka variance ban jaata hai) — notes me 0.004 sirf rounding se aaya hai (140/3 = 46.666... ko 46.66 likha). Isliye ye tarika split compare karne ke liye sahi nahi hai. Exam me apne teacher se confirm kar lena ki kaunsa tareeka likhna hai.

Sahi calculation (child ke apne mean ke saath):

Split	Child	Values	Apna mean	Variance
A: Exp ≤ 2	C_1	[40]	40	0
	C_2	[42, 52, 60, 56]	52.5	(110.25 + 0.25 + 56.25 + 12.25) / 4 = 44.75
B: Exp ≤ 2.5	C_1	[40, 42]	41	(1 + 1) / 2 = 1
	C_2	[52, 60, 56]	56	(16 + 16 + 0) / 3 = 10.67

$$\mathbf{VR(A) = 60.8 - [\ (1/5) \times 0 + (4/5) \times 44.75\ ] = 60.8 - 35.8 = 25.0}$$
$$\mathbf{VR(B) = 60.8 - [\ (2/5) \times 1 + (3/5) \times 10.67\ ] = 60.8 - (0.4 + 6.4) = 54.0}$$

Yaad rakhein: Sahi calculation me bhi **Exp ≤ 2.5 jeetta hai** (54.0 > 25.0), bas ab fark saaf dikhta hai. Isliye teacher ka final tree aur answer bilkul sahi hai; sirf variance nikalne ka tareeka sudharna tha.

6. Final Tree Aur Prediction (Test Data)

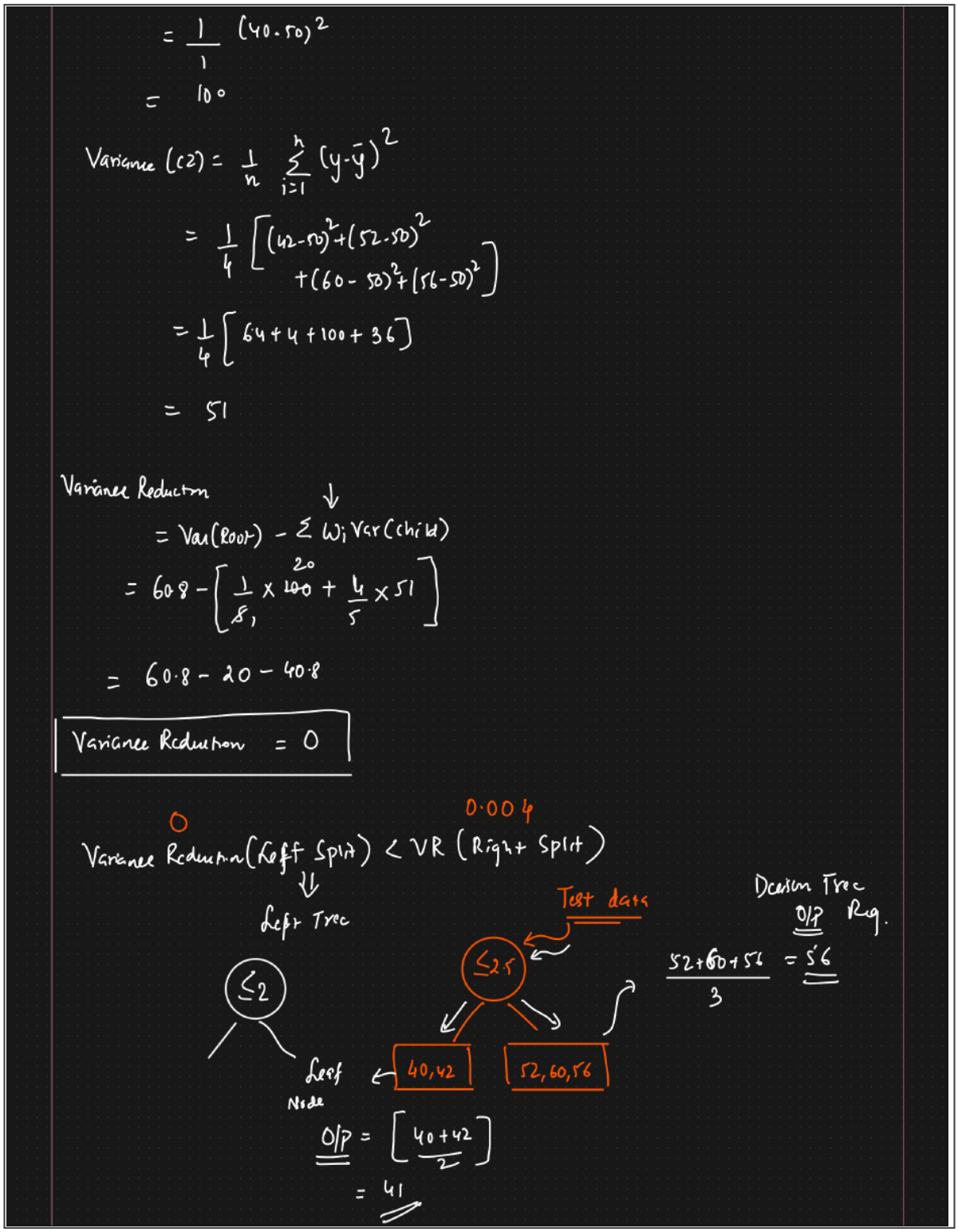
Best split **Exp ≤ 2.5** ban gaya. Do leaf nodes mile: **[40K, 42K]** aur **[52K, 60K, 56K]**. Regression tree me **leaf ka output = us leaf ke saare values ka average** hota hai (classification me majority class hoti thi):

Leaf	Condition	Values	Output (average)
Left leaf	Exp ≤ 2.5	40, 42	(40 + 42) / 2 = **41K**
Right leaf	Exp > 2.5	52, 60, 56	(52 + 60 + 56) / 3 = **56K**

Test data example: naya candidate jiska Exp = 2.2 saal hai. Root par sawal: 2.2 ≤ 2.5? **Yes** → left leaf → predicted salary = **41K**. Agar Exp = 3.5 hota, to 3.5 ≤ 2.5 **No** → right leaf → predicted salary = **56K**.

Dhyaan dein: Ek leaf me girne wale saare test points ko **same value** milti hai (yahan 41K ya 56K). Isliye regression tree ki prediction **seedhi-seedhi steps jaisi** hoti hai, smooth line jaisi nahi. Tree jitna gehra, utne zyada steps aur utna fine fit.

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Isme dikhaya gaya hai: C1 aur C2 ke variance ki detail calculation, Variance Reduction (Exp ≤ 2) = 0, dono splits ka comparison (0 < 0.004), aur test data ke liye leaf ka output: (40 + 42) / 2 = 41 aur (52 + 60 + 56) / 3 = 56.

7. Bonus — Saare Splits Ka Sahi Variance Reduction

(Ye section teacher ke notes me nahi hai, maine alag se joda hai.)

Teacher ne sirf do splits dekhe the. Dekhte hain agar saare candidate splits (sahi tareeke se, child ke apne mean ke saath) check karein to kaun sa best nikalta hai:

Split	Left | Right	Variance Reduction
Exp ≤ 2	[40] | [42, 52, 60, 56]	25.0
Exp ≤ 2.5	[40, 42] | [52, 60, 56]	**54.0 (best)**
Exp ≤ 3	[40, 42, 52] | [60, 56]	42.7
Exp ≤ 4	[40, 42, 52, 60] | [56]	9.0
Career Gap (Yes | No)	[40, 42, 56] | [52, 60]	24.0

Sabse zyada reduction **Exp ≤ 2.5** ka hai (54.0), yaani teacher ne bilkul sahi split chuna tha. Ye bhi dikhta hai ki Career Gap feature (24.0) is level par Exp ≤ 2.5 se kamzor hai. Aage Left aur Right leaf nodes ko bhi isi tarah todte ja sakte hain, jab tak variance bahut kam na ho jaaye ya tree ki limit na aa jaaye.

Practical baatein:

- Scikit-learn me **criterion = "squared_error"** default hai, jo yahi variance (MSE) wala tareeka hai. Doosre options: "friedman_mse", "absolute_error", "poisson".
- Classification tree ki tarah regression tree bhi **overfit** hota hai; ise rokne ke liye **pre-pruning** (max_depth, min_samples_leaf) ya **post-pruning** (ccp_alpha) wahi tarike lagte hain jo Decision Tree Classifier ke notes me hain.
- Numerical feature par threshold chunne ka tareeka (sort karo, har threshold try karo) bilkul wahi hai jo "Numerical Features Par Split" section me tha.

```
from sklearn.tree import DecisionTreeRegressor
model = DecisionTreeRegressor(max_depth=3, min_samples_leaf=2)
model.fit(X_train, y_train)
```

8. Poore Notes Ka Quick Revision

Topic	Yaad rakhne wali baat
Decision Tree Regressor	Output number ho tab use hota hai; algorithm CART
Purity measure	Variance = $(1/n)\ \Sigma\ (y - \bar{y})^2$ (yahi MSE hai); kam variance = pure
Best split	VR = Var(Root) − $\Sigma\ w_i \times$ Var($Child_i$); sabse bada VR chuno
Child ka variance	Hamesha child ke apne mean se nikalo (root ke mean se nahi)
Leaf ka output	Leaf ke saare values ka average (jaise (40 + 42) / 2 = 41)
Notes ka example	Root variance 60.8; best split Exp ≤ 2.5; outputs 41K aur 56K
Classification vs Regression	Entropy/Gini + Info Gain + majority class vs Variance + Variance Reduction + average